VismellNav: Learning Vision-Olfaction Fusion for Robust Odor Source Navigation

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Abstract

Vision-based navigation has achieved strong performance in structured environments, yet it remains vulnerable to visual occlusion, degradation, and novel, unseen scenarios. Biological systems transcend these limitations by using olfactory cues to navigate toward and localize targets in visually compromised environments. To bridge this gap, we propose VismellNav, a quadruped navigation framework that integrates visual perception with distributed olfactory sensing to achieve robust, target-oriented navigation in unseen environments. To synthesize these heterogeneous streams, we introduce a gated fusion mechanism within a flow-matching policy that assigns observation-dependent weights to visual and olfactory features. We evaluate VismellNav in complex indoor mazes where targets are visually occluded, requiring the robot to navigate around obstacles to localize the odor source. Experimental results demonstrate that our fusion policy significantly outperforms vision-only and olfaction-only baselines in unseen scenarios, achieving a 45.7% improvement in success rate over the vision-only approach for odor-source navigation. These findings highlight the role of cross-modal synergy in overcoming perception gaps to enable robust navigation in complex, real-world environments.

1 Introduction

Vision-based navigation has become a central capability for embodied robots, enabling agents to perceive scene structure, avoid obstacles, and make goal-directed decisions from egocentric observations [1–4]. Recent advances in visual representation learning, imitation learning, and foundation-style navigation policies have further improved robot navigation in structured indoor environments [5–9]. However, most vision-based policies still depend on reliable visual evidence; when the target is occluded, observations are degraded, or test environments differ from training scenes, vision-only policies often struggle to infer the goal location [10–12]. This limitation is especially important in odor-source navigation, a long-standing problem in mobile robot olfaction [13–16]. The odor source may be physically present but visually inaccessible due to obstacles, walls, or local turns. Although vision provides local geometry and free-space information, it may lack target-related cues for judging whether the robot is moving closer to the source. Biological studies suggest that non-visual cues, including olfaction, can help overcome such perceptual gaps: humans can track odor trails and use spatial differences in olfactory signals to support navigation [17, 18]. Unlike vision, olfaction does not require direct visibility of the source, and can provide intermittent but target-related evidence when visual cues are incomplete. Prior work in robotic olfaction has studied gas-source localization, plume tracking, and chemically guided search with mobile robots [13,14,19–22]. Other studies combine olfactory and visual sensing for odor-source localization, while broader navigation work explores multi-modal fusion for perception and control [23–28]. These works highlight the potential of olfaction in robot navigation, but often rely on olfaction-dominant strategies, simplified settings, or non-legged platforms. How legged robots should fuse vision, distributed chemical cues, and proprioception for odor-source navigation in visually unfamiliar scenes remains underexplored. To address this problem, we propose VismellNav, a vision–olfaction navigation framework for quadruped robots searching for hidden odor sources in visually occlude indoor environments. VismellNav equips a Unitree Go2 robot with an RGB camera, four leg-mounted electronic nose modules, and proprioceptive sensing, allowing the policy to jointly capture scene structure, distributed chemical cues, and robot motion states. Rather than treating olfaction as a standalone directional signal, VismellNav learns a unified condition representation for closed-loop action generation. Specifically, visual features are extracted by a frozen DINOv2 ViT-B/14 encoder and processed by a causal temporal Transformer for historical and goal-aware visual modeling. Proprioceptive states and multi-channel olfactory measurements are encoded by lightweight MLPs. These features are integrated through an adaptive gated fusion module, which predicts modality-specific weights according to the current observation context.

The fused condition is then used by a conditional U-Net to

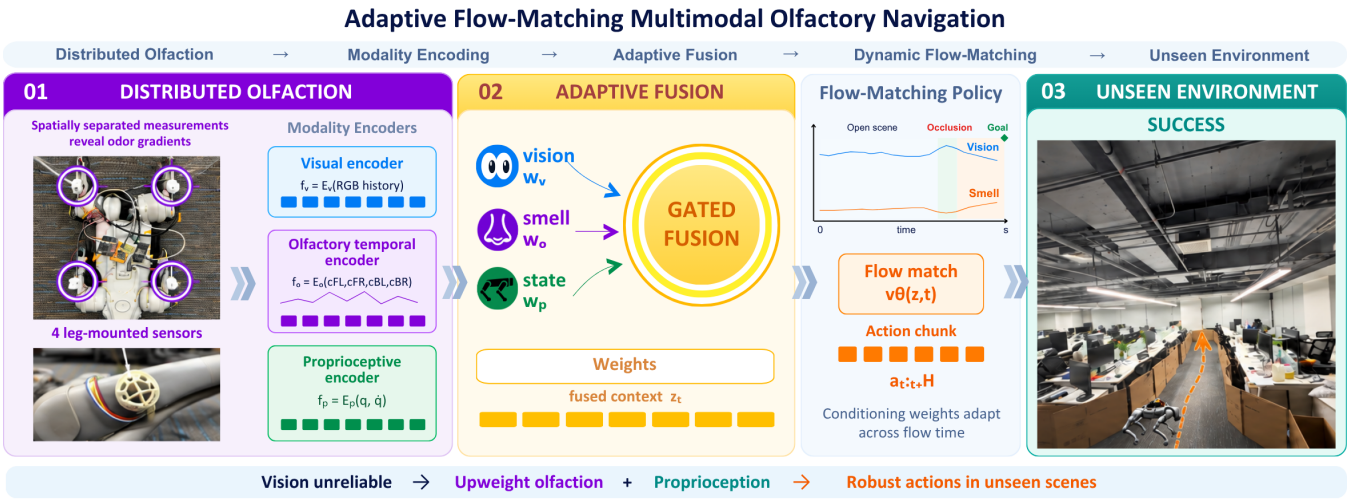


Figure 1: **Overview of the proposed adaptive flow-matching framework for multimodal olfactory navigation.** Spatially distributed olfactory sensors mounted on the robot's four legs capture spatial variations in odor responses, while visual, olfactory, and proprioceptive observations are encoded separately. A gated fusion module dynamically reweights the three modalities to produce a fused context for the flow-matching policy, which generates action chunks for navigation.

predict the velocity field of future action trajectories under a flow-matching formulation [29]. At inference time, Euler integration generates the action sequence, and a vision-based goal head provides an automatic stopping signal when the target is consistently detected. We evaluate VismellNav on real-world odor-source seeking in maze environments. During evaluation, we construct four cardboard maze layouts that are not used during training, introducing target occlusion, partial observability, and visually unfamiliar layouts. Experimental results show that VismellNav consistently outperforms vision-only and olfaction-only baselines.

The olfaction-only baseline fails across all tested environments, suggesting that olfactory sensing alone is insufficient for robust navigation, but becomes effective when fused with visual and proprioceptive observations.

2 Related Works

2.1 Vision-based Navigation

Vision-based navigation is a core problem in embodied intelligence and mobile robotics [30–32]. Early learning-based methods map egocentric visual observations to goal-directed actions, enabling indoor navigation from RGB inputs [33]. Later studies introduce spatial memory, topological representations, and semantic reasoning for long-horizon navigation and exploration [1–3, 34, 35]. Recent work further improves generalization across environments, robots, and tasks through large-scale datasets, foundation-style navigation policies, and diffusion-based action generation [6–8, 36].

Vision Language and Foundation Model Navigation

Beyond image-goal navigation, vision-language and foundation-model navigation have become important directions for high-level generalization. These models allow agents to interpret semantic instructions, reason about goals, and perform zero-shot navigation in unexplored environments [9,37,38]. Recent methods further explore unified navigation models, grounded vision-language reasoning, and improved waypoint prediction for generalist navigation [39–41]. Despite this progress, most systems still rely on visual appearance or semantic grounding, making them vulnerable when the target is ambiguous, occluded, outside the field of view, or located in unseen environments. Different from these appearance-driven paradigms, our work studies vision–olfaction fusion, where distributed olfactory sensing provides spatial guidance beyond visual observations.

2.2 Olfaction-based Navigation

Olfaction has long served as a cue for odor or gas source localization, especially under visual occlusion or visually unseen conditions [13,14,19,20]. Human scent-tracking studies show that spatial odor concentration differences can provide directional cues for source seeking [17]. In robotics, odor source localization has been studied through plume tracking, swarm intelligence, and multi-robot collaboration [15]. Cooperative robots can improve search robustness in turbulent and patchy odor fields [15, 42]. Recent learning-based methods further extend olfactory navigation, including diffusion graph neural networks for robust olfactory navigation and hazard robotics [22], sim-to-real olfactory navigation stacks [43], and olfactory inertial odometry [21]. Robotic chemical sensing systems, such as portable mass spectrometry, also show the potential of mobile platforms for on-site detection of hazardous volatile compounds [44]. Different from olfaction-centered source localization or chemical detection, our work uses distributed olfactory sensing as a spatial guidance signal and fuses it with visual observations to improve navigation robustness when visual cues are occluded, miss-

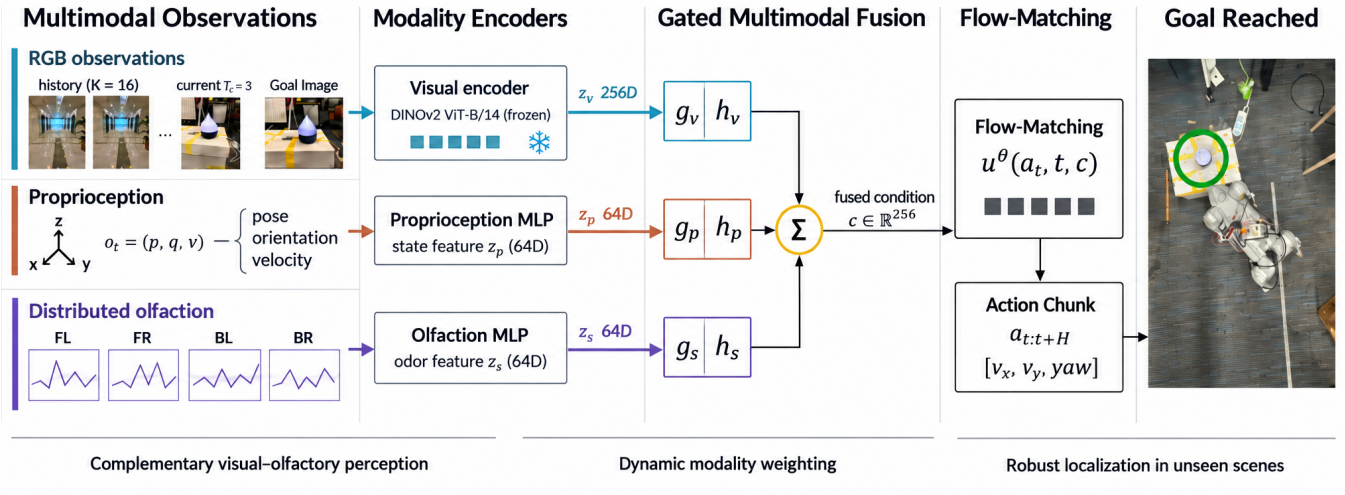


Figure 2: **Overview of the VismellNav multimodal navigation pipeline.** RGB observations, proprioceptive states, and distributed olfactory measurements are encoded into modality-specific features and adaptively combined through gated multimodal fusion. The fused representation conditions a flow-matching policy that predicts action chunks for goal-directed navigation, enabling dynamic modality weighting and robust odor-source localization under varying visual conditions.

ing, or poorly generalized to unseen scenes.

2.3 Multi-modality Fusion

Multi-modal fusion improves robotic perception and navigation under noisy, partially observable, and visually degraded conditions [25–27, 45]. By combining complementary cues such as vision, LiDAR, IMU, tactile sensing, and chemical sensing, robots achieve more reliable state estimation and decision-making than with a single modality [28, 46]. Recent work further shows that modality importance varies with environmental context, motivating adaptive fusion mechanisms instead of fixed feature concatenation [47]. In robotic olfaction, prior work studies vision–olfaction fusion for gas and odor source localization [23, 24, 48], while other systems combine heterogeneous cues such as olfaction and hearing for source-seeking tasks [49]. However, most existing approaches remain focused on explicit odor source localization using handcrafted strategies or high-level reasoning.

3 Method

3.1 Problem Formulation

We consider goal-conditioned odor-source navigation in indoor environments containing a single active odor source. Given a reference image of the target object containing the source, a quadruped robot must navigate to the odor-bearing target while avoiding obstacles. The robot uses egocentric visual observations, distributed olfactory measurements, and proprioceptive states to guide navigation, including when the target is visually occluded.

At each time step t, the policy receives a goal-conditioned multi-modal observation

$$\mathcal{O}_t = \{I_t, I_g, X_{t-k+1:t}\}, \tag{1}$$

where $I_t \in \mathbb{R}^{3\times224\times224}$ denotes the current RGB observation from the onboard camera and $I_g \in \mathbb{R}^{3\times224\times224}$ denotes a goal image of the target object containing the odor source, provided before deployment. The goal image remains fixed throughout an episode. The proprioceptive–olfactory history $X_{t-k+1:t} \in \mathbb{R}^{k\times20}$ contains the corresponding signals from the most recent k time steps. We use $k = 5$ in our implementation. Each history entry is represented as $X_\tau = [p_\tau, s_\tau]$, where $p_\tau = [\Delta x_\tau, \Delta y_\tau, \sin(\theta_\tau), \cos(\theta_\tau)] \in \mathbb{R}^4$ encodes the robot's planar displacement and heading, and $s_\tau \in \mathbb{R}^{16}$ contains the measurements from four leg-mounted electronic-nose modules.

At each time step, the policy outputs a continuous velocity command

$$\mathbf{a}_t = \begin{bmatrix} v_{x,t} & v_{y,t} & \omega_t \end{bmatrix}^\top, \tag{2}$$

where $v_{x,t}$, $v_{y,t}$, and ω_t denote the forward linear velocity, lateral linear velocity, and yaw angular velocity, respectively.

3.2 Adaptive Multimodal Fusion

Multi-modal observation encoding. As illustrated in Fig. 2, VismellNav separately encodes the RGB observations, proprioceptive states, and distributed olfactory measurements into modality-specific latent representations. The RGB observations and goal image are processed using a shared frozen DINOv2 visual encoder. The resulting visual features are then aggregated using a causal Transformer [7] to model recent visual context and its relationship to the goal, producing the visual representation $\mathbf{z}_v$. In parallel, the proprioceptive and olfactory histories are encoded using separate MLP en-

coders:

$$\mathbf{z}_p = f_p(\mathbf{p}_{t-k+1:t}), \qquad \mathbf{z}_s = f_s(\mathbf{s}_{t-k+1:t}), \tag{3}$$

where $\mathbf{z}_p$ and $\mathbf{z}_s$ denote the proprioceptive and olfactory representations, respectively. The multimodal observation encoder therefore produces three modality-specific representations: $\mathbf{z}_v$, $\mathbf{z}_p$, and $\mathbf{z}_s$.

Vision-olfaction gated fusion. The modality-specific representations are then passed to a gating network that predicts observation-dependent weights for the three modalities:

$$\mathbf{g} = \begin{bmatrix} g_v \\ g_p \\ g_s \end{bmatrix} = \mathrm{softmax}\left(W_g \begin{bmatrix} \mathbf{z}_v \\ \mathbf{z}_p \\ \mathbf{z}_s \end{bmatrix} \right), \tag{4}$$

where g_v, g_p, and g_s are scalar gating weights for vision, proprioception, and olfaction, respectively. The softmax is applied across thethree modalities, such that the weights are nonnegative and sum to one. After projecting the three representations into a shared latent space, the fused condition is computed as

$$\mathbf{c} = g_v h_v(\mathbf{z}_v) + g_p h_p(\mathbf{z}_p) + g_s h_s(\mathbf{z}_s), \tag{5}$$

where h_v, h_p, and h_s denote the modality-specific projection functions.

The observation-dependent gating allows the policy to adjust its reliance on each modality as the navigation context changes. When the target is visually occluded or difficult to localize from RGB observations alone, olfactory measurements can provide complementary target-related cues. Conversely, visual observations provide geometric information for obstacle avoidance and local navigation, while proprioception captures the robot's current motion state. The resulting multimodal condition $\mathbf{c}$ is used for both action generation and goal-reaching prediction.

3.3 Flow-Matching Navigation Policy

Flow-matching action generation. Conditioned on the multimodal representation $\mathbf{c}$, the policy generates future action trajectories using conditional flow matching. Rather than directly regressing actions, it learns a velocity field that transports Gaussian noise toward expert action trajectories. During training, an expert trajectory $\mathbf{x}_0^* \in \mathbb{R}^{T_p \times D_a}$, Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and an interpolation time $\tau \sim \mathcal{U}(0, 1)$ are sampled to construct $\mathbf{x}_\tau = (1-\tau)\boldsymbol{\epsilon} + \tau \mathbf{x}_0^*$, which evolves from Gaussian noise at $\tau = 0$ to the expert trajectory at $\tau = 1$. The corresponding target velocity is $\mathbf{v}^* = \mathbf{x}_0^* - \boldsymbol{\epsilon}$. A conditional 1D U-Net then predicts the velocity field as $\hat{\mathbf{v}}_\theta = v_\theta(\mathbf{x}_\tau, \tau, \mathbf{c})$. The network is optimized using the flow-matching objective

$$\mathcal{L}_{\text{flow}} = \|\hat{\mathbf{v}}_\theta - \mathbf{v}^*\|_2^2 . \tag{6}$$

During deployment, the policy initializes an action trajectory from Gaussian noise and numerically integrates the learned velocity field from $\tau = 0$ to $\tau = 1$, conditioned on the current multimodal representation $\mathbf{c}$. Only the first T_a actions of the generated trajectory are executed before the policy replans using updated observations. This receding-horizon execution allows the robot to adapt its actions as the visual and olfactory observations change during navigation.

Goal-reaching prediction. In addition to action generation, the multimodal representation $\mathbf{c}$ is used to predict whether the robot has reached the odor source. A lightweight goal-reaching head outputs a logit ℓ_{loc} and the corresponding probability $\hat{p}_{\text{loc}}$:

$$\ell_{\text{loc}} = f_{\text{loc}}(\mathbf{c}), \qquad \hat{p}_{\text{loc}} = \sigma(\ell_{\text{loc}}). \tag{7}$$

The ground-truth label y_{loc} is set to one when the target object is visible in the onboard camera view and the robot is within $0.4\,\text{m}$ of the odor source, and to zero otherwise. The goal-reaching head is trained jointly with the navigation policy using

$$\mathcal{L}_{\text{loc}} = \mathrm{BCEWithLogits}\left(\ell_{\text{loc}}, y_{\text{loc}}\right). \tag{8}$$

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{flow}} + \lambda_{\text{loc}} \mathcal{L}_{\text{loc}}. \tag{9}$$

During deployment, the robot continuously monitors $\hat{p}_{\text{loc}}$ while executing navigation actions. Navigation is terminated when $\hat{p}_{\text{loc}}$ remains above a predefined threshold for a prescribed number of consecutive replanning steps.

4 Experiments

4.1 Robot Platform and Olfactory Sensing

Robot Platform. We use a Unitree Go2 EDU quadruped robot as the mobile platform for the real-world vision–olfaction navigation experiments. Visual observations are captured by the robot's onboard RGB camera. Four Mini5 electronic-nose modules are mounted on the robot, with one module attached to each leg to provide spatially distributed olfactory measurements. A Raspberry Pi mounted on the robot's back handles olfactory data acquisition.

Electronic Nose. Each Mini5 contains four cross-sensitive metal-oxide-semiconductor (MOS) gas-sensing channels based on distinct sensing materials: S1 ($\mathrm{Pd/SnO_2}$), S2 ($\mathrm{Pd/WO_3}$), S3 ($\mathrm{Pt/SnO_2}$), and S4 ($\mathrm{Pt/WO_3}$) [50]. Previous characterization demonstrated differentiated responses to several volatile compounds, including acetic acid, ethanol, methanol, acetone, toluene, and ethyl acetate [50]. The response to acetic acid is particularly relevant because white vinegar is used as the primary odor source. Together, the four Mini5 modules provide 16 olfactory channels sampled at approximately 10 Hz and synchronized with the RGB observations and proprioceptive states. VismellNav directly uses these multi-channel responses as olfactory inputs, without explicitly classifying the odor or estimating its absolute concentration.

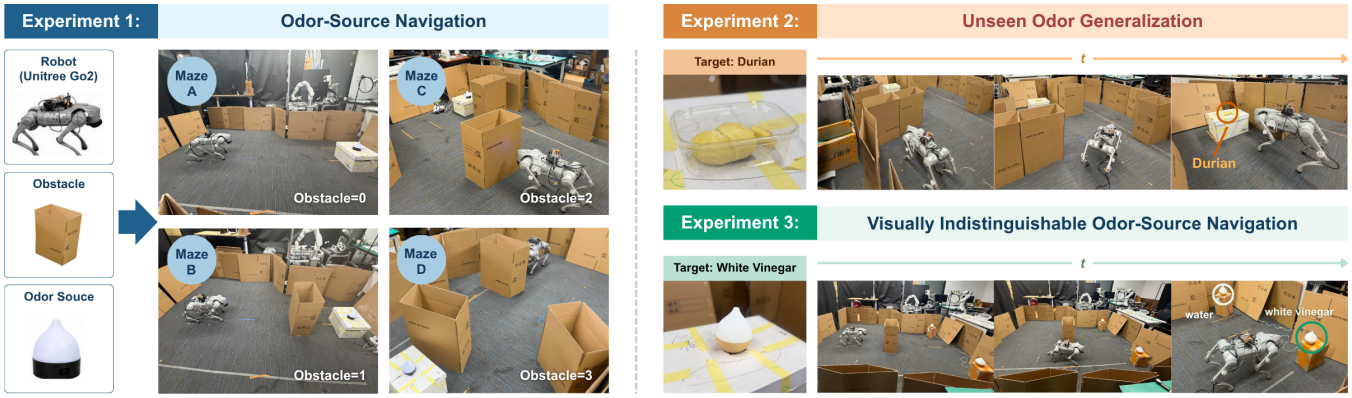


Figure 3: **Real-world experimental setups for evaluating VismellNav.** Experiment 1 evaluates odor-source navigation across four maze layouts with different obstacle configurations. Experiment 2 tests generalization to an unseen odor source, durian. Experiment 3 evaluates target selection in Maze A and Maze B when the candidate odor sources are visually indistinguishable.

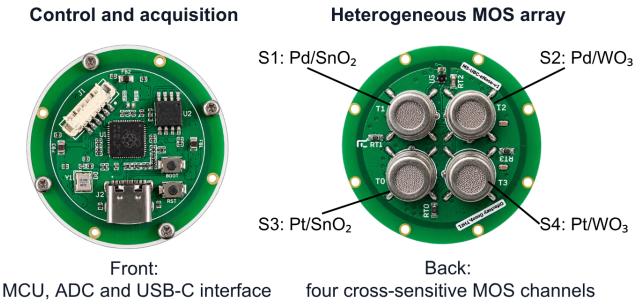


Figure 4: **The Mini5 electronic-nose module and its four gas-sensing channels used in VismellNav.** The front side integrates the MCU, ADC, and USB-C interface for sensor control and data acquisition, while the back side contains a heterogeneous array of four cross-sensitive MOS sensors with different sensing materials (Pd/SnO_2, Pd/WO_3, Pt/SnO_2, and Pt/WO_3). Their complementary responses form a multichannel odor fingerprint for downstream olfactory perception.

4.2 Experimental Setup

Maze Environments and Odor-Source Preparation. All experiments were conducted in real-world indoor maze environments constructed from cardboard boxes, as shown in Fig. 3. The boxes defined the navigable area and created different obstacle and target-occlusion configurations. The initial source-to-robot distance was maintained at approximately 2.0–2.5 m across the evaluated maze configurations. No fans or other sources of artificially imposed directional airflow were used during the experiments.

For trials using white vinegar, the odor solution was prepared by mixing 80% (v/v) white vinegar with 20% water. For each trial, 80 mL of the prepared solution was loaded into an ultrasonic diffuser placed at the target location and released at a nominal rate of 16 mL/h. The diffuser was activated 15 min before robot initialization to allow the odor to disperse and remained active throughout the episode.

After each diffuser-based trial, all active diffusers were switched off, and the environment was ventilated for 10 min to reduce residual odor accumulation. The Mini5 modules were then allowed to recover in ambient air for an additional 5 min. Before the first trial of each experimental session, all Mini5 modules were warmed up for 15 min to stabilize their baseline responses.

Baselines and Evaluation Metrics. In Experiment 1, we compare VismellNav with four baselines: Vision-only, Olfaction-only, Feature Concatenation, and Fixed-weight Fusion. Vision-only and Olfaction-only remove the olfactory and visual observations, respectively, while the latter two use all three modalities but replace adaptive gated fusion with non-adaptive fusion strategies. For Experiments 2 and 3, we compare VismellNav with the Vision-only and Olfaction-only baselines to focus on odor transfer and target disambiguation.

Each evaluated method is tested over 20 trials under each experimental condition. We report success rate as the primary evaluation metric. A trial is considered successful when the target object containing the odor source is visible in the robot's onboard RGB camera and the robot is within 0.4 m of the source. Each episode has a maximum duration of 180 s.

4.3 Results and Analysis

Experiment 1: Odor-Source Navigation and Fusion Ablation. We first evaluate VismellNav in real-world odor-source navigation tasks across four maze configurations. As shown in Fig. 3, Mazes A–D contain zero, one, two, and three obstacles along the navigation paths, respectively. These obstacles introduce different degrees of target occlusion and partial observability.

In addition to the Vision-only and Olfaction-only baselines, we evaluate one non-adaptive multimodal baseline to isolate the contribution of the proposed gated fusion mechanism. Fixed-weight Fusion combines the projected modality-specific features using fixed weights. This baseline receives the same multimodal inputs as VismellNav but replaces its adaptive gated fusion module with a non-adaptive fusion

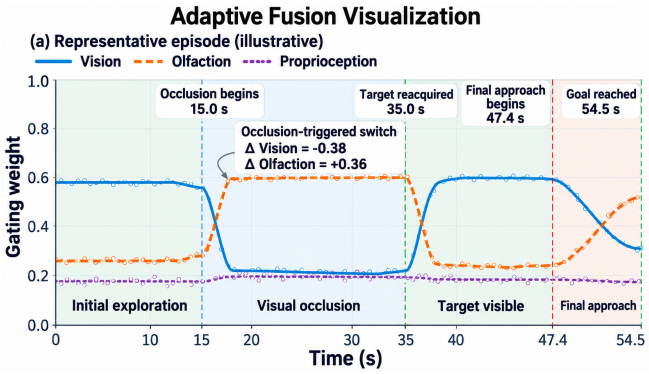


Figure 5: **Adaptive multimodal fusion during a representative navigation episode.** Learned gating weights dynamically shift across navigation phases: visual occlusion at 15.0 s suppresses vision and increases olfactory reliance, while target reacquisition at 35.0 s restores visual dominance. During the final approach from 47.4 s, olfactory weighting increases again near the odor source.

strategy.

As shown in Fig. 6, VismellNav achieves the highest success rates in all four maze configurations, reaching 85%, 60%, 55%, and 55% in Mazes A–D, respectively. The Vision-only baseline achieves 60%, 55%, 25%, and 35%, while Olfaction-only achieve 0% in all four mazes. Fixed-weight Fusion succeeds only in Maze A, with a success rate of 20%.

Averaged across the four maze configurations, VismellNav achieves a success rate of 63.8%, compared with 43.8% for Vision-only. This corresponds to a 20.0 percentage-point absolute improvement, or a 45.7% relative improvement. These results show that adaptive multimodal fusion improves navigation robustness across unseen maze layouts.

Experiment 2: Generalization to an Unseen Odor. We evaluate whether VismellNav can navigate toward an odor source not encountered during training. Durian is selected as a naturally emitting odor source that is qualitatively different from the vinegar-based source used during training. As shown in Fig. 3, a container of durian is placed at the target location, and the policies are evaluated in maze environments containing obstacles and target occlusion. No odor-specific adaptation is introduced for this evaluation.

As summarized in Fig. 7, VismellNav achieves a success rate of 40% when navigating toward the unseen durian odor, compared with 25% for Vision-only and 0% for Olfaction-only. This result indicates that the learned multimodal policy can transfer beyond the vinegar-based odor condition used during training. Nevertheless, because only one unseen odor is evaluated, the result demonstrates transfer to the tested durian odor rather than broad generalization across arbitrary odor sources.

Experiment 3: Target Selection under Visual Ambiguity. We evaluate whether olfactory sensing can resolve target ambiguity when visual appearance alone cannot identify the odor-bearing target. The two evaluation settings use the Maze A and Maze B layouts from Experiment 1, respectively. As shown in Fig. 3, two visually identical ultrasonic diffusers are placed at candidate target locations, with one containing white vinegar and the other containing water. Obstacles are positioned in the environment to introduce target occlusion and partial observability.

A trial is counted as successful only when the robot reaches the vinegar-containing diffuser according to the success criterion defined above. Approaching the water-containing diffuser is considered a failure.

As summarized in Fig. 7, VismellNav achieves success rates of 100% and 65% in Mazes A (zero obstacles) and B (one obstacle), respectively, compared with 35% and 25% for Vision-only. Olfaction-only achieves 0% in both mazes. These results indicate that olfactory observations provide discriminative information for identifying the correct target when the candidate sources cannot be distinguished by visual appearance alone.

5 Limitations

Our real-world evaluation covers a limited set of indoor maze environments with relatively controlled and static obstacle configurations. Although Experiment 2 evaluates transfer to one unseen odor, the results do not establish broad generalization across diverse odor sources, source strengths, airflow conditions, temperatures, or humidity levels. In addition, MOS gas sensors exhibit response delay, signal drift, cross-sensitivity, and slow recovery. VismellNav learns directly from distributed temporal measurements but does not explicitly model these sensor dynamics or the underlying odor transport process.

The contribution of individual sensor locations and the necessity of the four-module configuration have not yet been systematically quantified through sensor-removal or corruption studies. Obstacle avoidance and long-horizon stability also remain to be evaluated in larger, more cluttered, and dynamic environments. Future work will extend the evaluation to more diverse environments and odor sources, incorporate environmental and sensor-dynamic modeling, and improve collision-aware action generation.

6 Conclusion

We presented VismellNav, a real-world vision–olfaction navigation framework for quadruped odor source seeking in visually occluded and unseen indoor environments. VismellNav integrates ego-centric vision, distributed olfactory measurements, and proprioceptive states through adaptive gated fusion to condition flow-matching action generation. Real-world experiments show that the robot can locate visually hidden odor sources under target occlusion, with moderate gains in seen environments and clearer advantages in unseen maze layouts. These findings highlight the value of distributed olfactory sensing as complementary, appearance-independent guidance for navigation when visual observations are occluded, ambiguous, or poorly generalized.

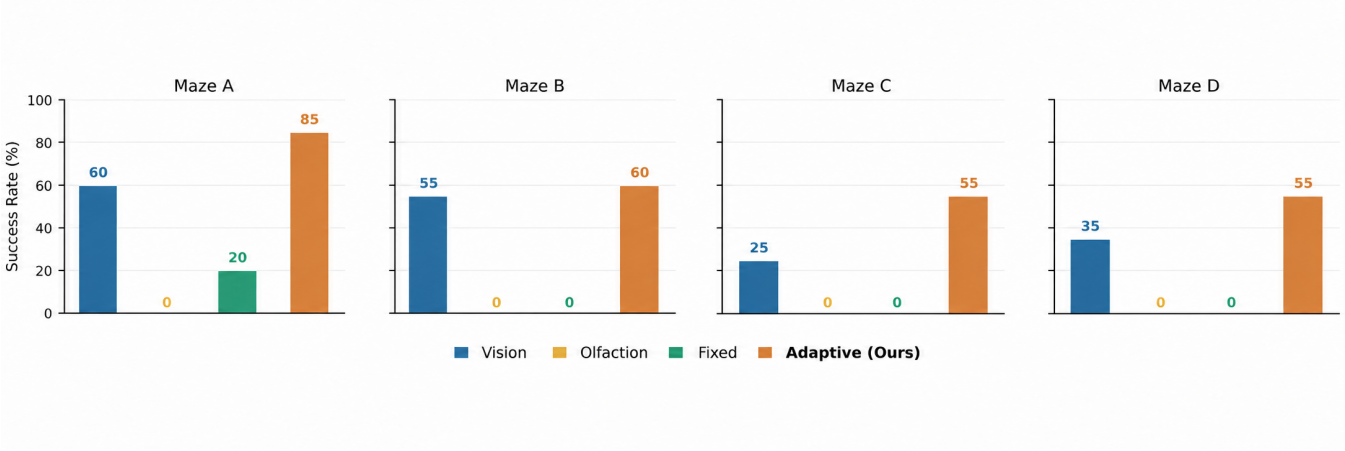


Figure 6: **Success rates for Experiment 1 (odor-source navigation and fusion ablation).** Each method is evaluated over 20 trials in each of the four real-world maze configurations (Mazes A–D, containing zero, one, two, and three obstacles, respectively). We compare the proposed adaptive gated fusion (VismellNav) against vision-only, olfaction-only, and fixed-weight fusion baselines. VismellNav achieves the highest success rate in all four mazes.

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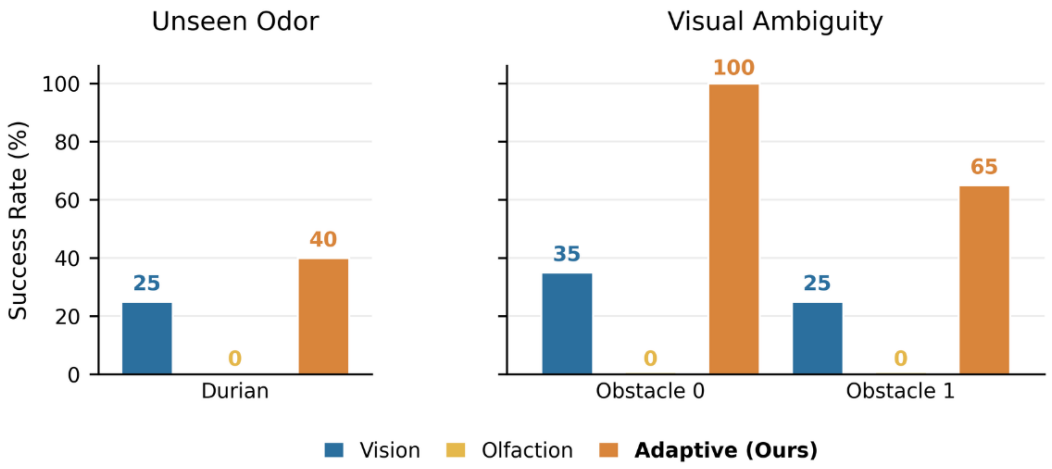


Figure 7: **Success rates across Experiments 2, 3.** From left to right: (Exp. 2) generalization to an unseen odor source (durian) not encountered during training; (Exp. 3) target selection under visual ambiguity, where two visually identical diffusers contain white vinegar and water, respectively, and only reaching the vinegar diffuser counts as success.

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